JC2001 Introduction to Software Engineering

Practical Session 5 - Exercises

(Model Answers)

Use Cases

Exercise A: Develop a top-level use case diagram for a Point-of-Sale (POS) system

Question 1: List and describe all the actors in a point-of-sale.
Answer: For a top-level use case diagram, the actors are:

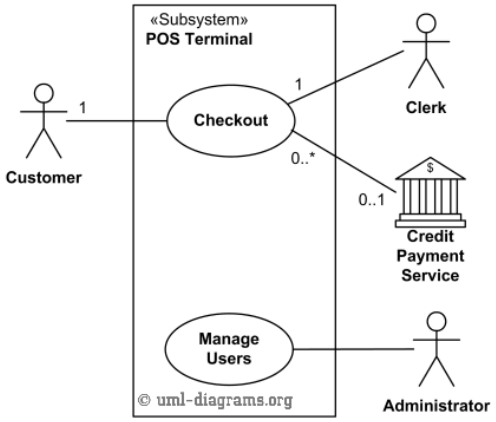
- **Customer:** The person who purchases goods and participates in the Checkout process.
- **Clerk:** The staff member who operates the POS terminal and carries the checkout process.
- **Credit Payment Service:** An external service involved in processing payment when required.
- **Administrator:** The person responsible for managing users of the POS system.

Question 2: List and describe all the processes involved in executing a sale.
Answer: The main process involved in executing a sale is Checkout**.** Manage users appears at the top-level diagram but is not part of a typical execution of a sale.

Question 3: Plot a use case diagram for a POS.
Answer:

An example of UML **use case diagram** for **Point of Sale (POS) Terminal** or Checkout. A retail POS system typically includes a computer, monitor, keyboard, barcode scanners, weight scale, receipt printer, credit card processing system, etc. and POS terminal software.

Checkout **use case** involves Customer, Clerk and Credit Payment Service **actors** and **includes** scanning items, calculating total and taxes, payment use cases.



Top level UML use cases for Point of Sales Terminal (POS).

Exercise B: Risk Analysis

Question 4: List and describe all the actors in the checkout use case.

Answer: The actors involved in the checkout use case are:

- **Customer:** participates in the checkout process.

- **Clerk**: carries out the checkout process. The sample solution indicates that one clerk participates in a single checkout use case.
- **Credit Payment Service**: May participate when the payment method requires a credit payment service.
  The sample solution explicitly explains that Credit Payment Service participation is optional because some payment methods, such as cash, do not require it.

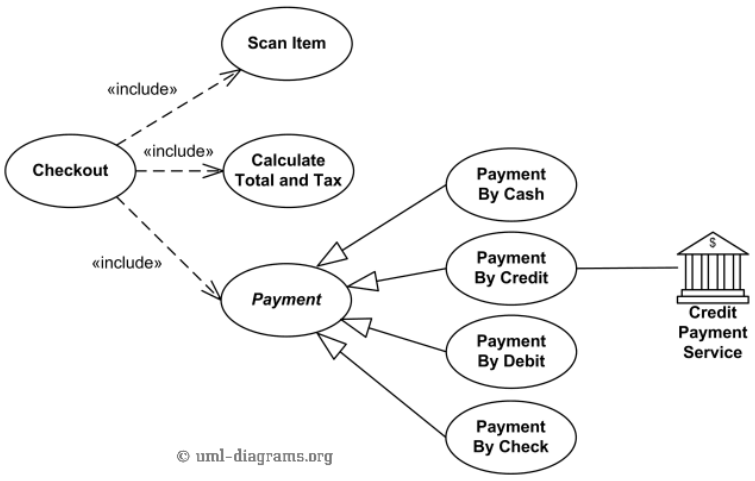
Question 5: List and describe all the processes in the checkout use case.
Answer: The checkout use case includes the following processes:

- **Scan Item:** Records the item being purchased.
- **Calculate Total and Tax:** Calculates the total purchase amount and applicable tax.
- **Payment:** Processes payment for the transaction.
  The payment use case is specialised into:
  - Payment by Cash
  - Payment by Credit
  - Payment by Debit
  - Payment by Check (Cheque)

Question 6: Plot a use case diagram for the checkout use case.
Answer:

Checkout use case requires Customer actor, hence the 1 multiplicity of Customer. Clerk can only participate in a single Checkout use case. Credit Payment Service can participate with many Checkout use cases at the same time. Checkout use case may not need Credit Payment Service (for example, if payment is in cash), thus the 0..1 multiplicity.

Checkout use case is an example of a large and complex use case split into several use cases each describing some logical unit of behavior. Note, that including use case becomes incomplete by itself and requires the included use cases to be complete.



Checkout use case includes Scan Item, Calculate Total and Tax, and Payment use cases.

Payment use case is represented using **generalization** relationship. It means that only one specific type of payment is accepted - either by cash, or by credit, debit, or with check. An alternative to such representation could be to use **include** relationship so that not just single but several forms of payment could be accepted from the same client during checkout.